Time-Resolved Asymmetry: Human-AI vs Human-Human vs AI-AI

How story-internal asymmetry evolves across early / middle / late phases

PENPAL Project

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Contents

1 Overview

This notebook tracks how within-dyad asymmetry evolves *across the story.* For each metric (valence, novelty-resonance slope, semantic-exploration AUC), we compute the per-story signed and unsigned slot-1 vs slot-2 difference within rolling thirds (Early / Middle / Late) of the story turn window, and test whether condition × phase interactions appear.

Two outcomes per metric:

- **Magnitude**: per-story $|A|$ in each phase.
- **Direction**: per-story signed slot-difference in each phase (with the identity-tied directionality reading: HA should keep a stable sign across phases; HH and AA should show no fixed sign).

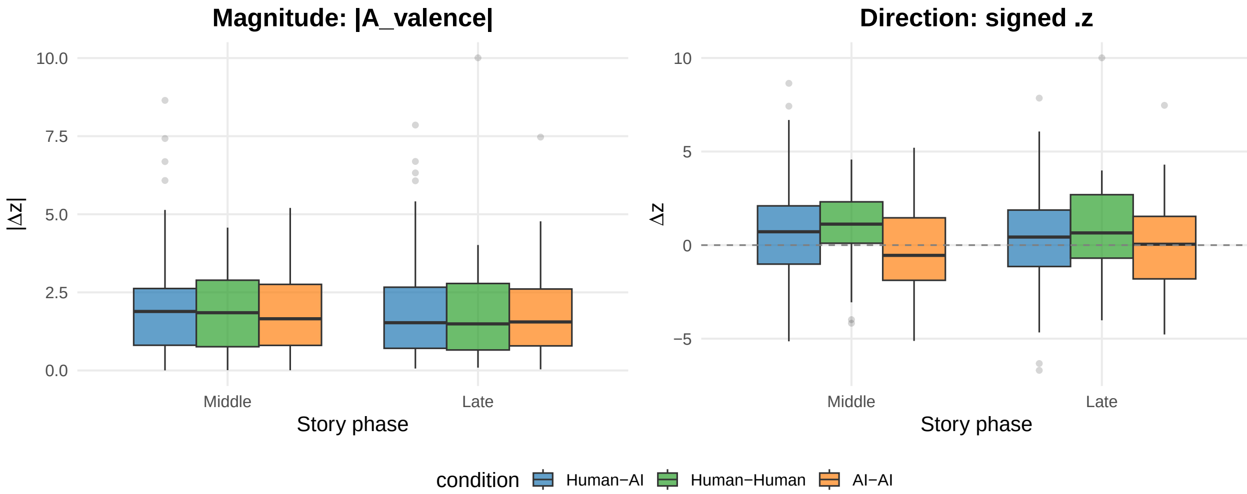
2 Phase definition

3 Valence Asymmetry by Phase

Table 1: Per-story valence asymmetry by condition and story phase

condition	phase	n	mean_signed	sd_signed	mean_abs
Human-AI	Middle	94	0.605	2.602	2.079
Human-AI	Late	98	0.340	2.518	1.938
Human-Human	Middle	36	0.859	2.247	1.982
Human-Human	Late	36	0.833	2.575	1.967
AI-AI	Middle	80	-0.348	2.269	1.875
AI-AI	Late	80	-0.003	2.309	1.835

condition	phase	n	mean_signed	sd_signed	mean_abs




```
## LMM: A_valence ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_valence ~ condition * phase + (1 | conversation_id)
##    Data: valence_phase_A
##
## REML criterion at convergence: 1568.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.3336 -0.7921 -0.1699  0.5113  5.1832
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.04718  0.2172
##  Residual                    2.30383  1.5178
## Number of obs: 424, groups:  conversation_id, 215
##
## Fixed effects:
##                              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                   2.07982    0.15815 417.86398  13.151   <2e-16
## conditionHuman-Human         -0.09744    0.30053 417.84063  -0.324    0.746
## conditionAI-AI               -0.20452    0.23323 417.84671  -0.877    0.381
## phaseLate                    -0.14174    0.21920 212.22232  -0.647    0.519
## conditionHuman-Human:phaseLate  0.12665    0.41957 207.38047   0.302    0.763
## conditionAI-AI:phaseLate      0.10142    0.32503 208.58119   0.312    0.755
##
## (Intercept)                 ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate
## conditionHuman-Human:phaseLate
## conditionAI-AI:phaseLate
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

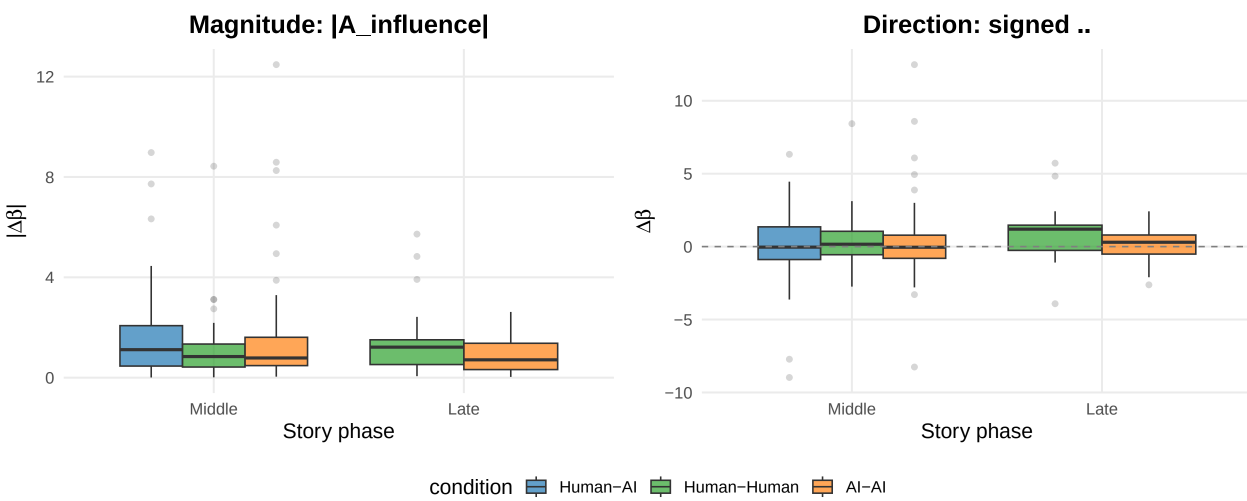
```
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H cnAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.526
## condtnAI-AI -0.678  0.357
## phaseLate   -0.708  0.372  0.480
## cndtnHm-H:L  0.370 -0.702 -0.251  -0.522
## cndtAI-AI:L  0.477 -0.251 -0.704  -0.674  0.352
##
## LMM: signed_diff ~ condition * phase + (1 | conversation_id)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: signed_diff ~ condition * phase + (1 | conversation_id)
##    Data: valence_phase_A
##
## REML criterion at convergence: 1955.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.8858 -0.6203  0.0188  0.6454  3.7655
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.000    0.000
##  Residual                    5.937    2.437
## Number of obs: 424, groups:  conversation_id, 215
##
## Fixed effects:
##                                Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                      0.6053     0.2513 418.0000   2.409   0.0164 *
## conditionHuman-Human             0.2542     0.4776 418.0000   0.532   0.5948
## conditionAI-AI                  -0.9536     0.3706 418.0000  -2.573   0.0104 *
## phaseLate                       -0.2653     0.3518 418.0000  -0.754   0.4512
## conditionHuman-Human:phaseLate   0.2386     0.6735 418.0000   0.354   0.7233
## conditionAI-AI:phaseLate         0.6101     0.5217 418.0000   1.170   0.2429
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H cnAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.526
## condtnAI-AI -0.678  0.357
## phaseLate   -0.714  0.376  0.484
## cndtnHm-H:L  0.373 -0.709 -0.253  -0.522
## cndtAI-AI:L  0.482 -0.254 -0.710  -0.674  0.352
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

4 Novelty-Resonance Slope Asymmetry by Phase

Table 2: Per-story novelty-resonance slope asymmetry by condition and story phase

condition	phase	n	mean_signed	sd_signed	mean_abs
Human-AI	Middle	95	0.123	2.198	1.513
Human-Human	Middle	36	0.447	1.869	1.200
Human-Human	Late	17	0.854	2.212	1.647
AI-AI	Middle	80	0.224	2.459	1.429
AI-AI	Late	39	0.173	1.180	0.939




```
## LMM: A_influence ~ condition * phase + (1 | conversation_id)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_influence ~ condition * phase + (1 | conversation_id)
##    Data: novelty_phase_A
##
## REML criterion at convergence: 1019.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.9721 -0.5446 -0.2435  0.1909  6.7680
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.000    0.000
##  Residual                    2.665    1.633
## Number of obs: 267, groups:  conversation_id, 211
##
## Fixed effects:
##                               Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                     1.5133     0.1675 262.0000   9.035   <2e-16
## conditionHuman-Human           -0.3135     0.3195 262.0000  -0.981    0.327
## conditionAI-AI                 -0.0845     0.2477 262.0000  -0.341    0.733
## phaseLate                      -0.4899     0.3188 262.0000  -1.537    0.126
## conditionHuman-Human:phaseLate  0.9368     0.5766 262.0000   1.625    0.105
##
```

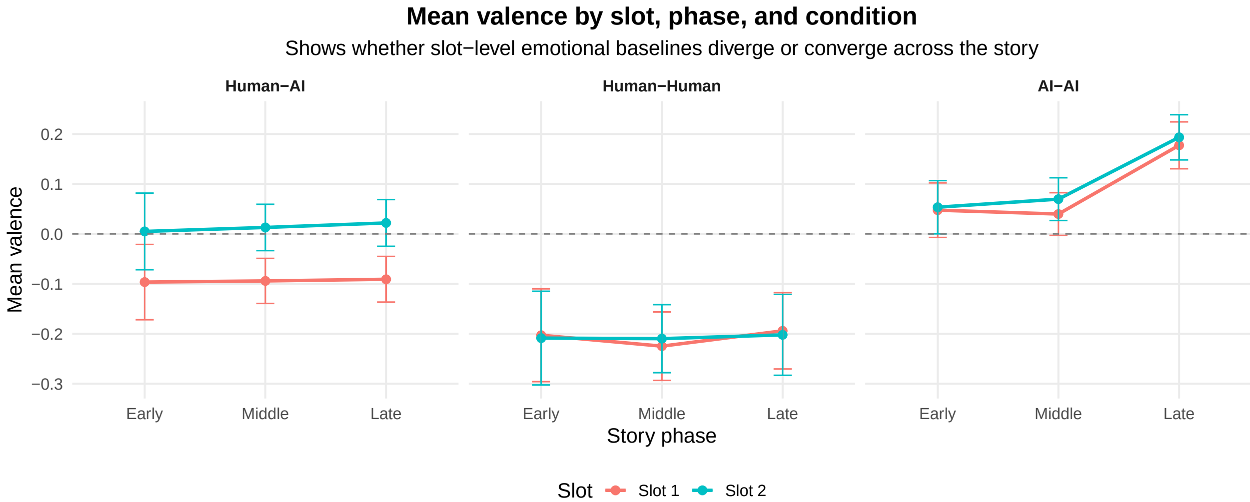
```
## (Intercept)                    ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate
## conditionHuman-Human:phaseLate
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H cAI-AI phasLt
## cndtnHmn-Hm -0.524
## condtnAI-AI -0.676  0.354
## phaseLate    0.000  0.000 -0.422
## cndtnHm-H:L  0.000 -0.402  0.233 -0.553
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
##
## LMM: signed_diff ~ condition * phase + (1 | conversation_id)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: signed_diff ~ condition * phase + (1 | conversation_id)
##    Data: novelty_phase_A
##
## REML criterion at convergence: 1158
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2766 -0.4619 -0.0747  0.3569  5.7614
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.000    0.000
##  Residual                    4.524    2.127
## Number of obs: 267, groups:  conversation_id, 211
##
## Fixed effects:
##                                 Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                      0.12280    0.21822 262.00000   0.563    0.574
## conditionHuman-Human             0.32438    0.41627 262.00000   0.779    0.437
## conditionAI-AI                   0.10082    0.32275 262.00000   0.312    0.755
## phaseLate                       -0.05018    0.41538 262.00000  -0.121    0.904
## conditionHuman-Human:phaseLate   0.45669    0.75120 262.00000   0.608    0.544
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H cAI-AI phasLt
## cndtnHmn-Hm -0.524
## condtnAI-AI -0.676  0.354
## phaseLate    0.000  0.000 -0.422
## cndtnHm-H:L  0.000 -0.402  0.233 -0.553
## fit warnings:
```

```
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

5 Mean Valence Trajectory by Slot and Condition

A descriptive companion view: how does each slot's mean valence drift across phase, by condition? Useful for the "story-evolution" narrative without relying on the asymmetry abstraction.



6 Summary

What this notebook adds beyond the cross-section comparisons:

- **Per-phase magnitude** $|A|$ for valence and novelty-resonance: tests whether asymmetry strengthens, weakens, or stays flat across the story.
- **Per-phase signed asymmetry**: combined with per-story directional-stability, this is the strongest visual evidence for HA-specific identity-tied directionality. A signed-asymmetry boxplot that stays *consistently above zero* in HA across all three phases, while HH/AA hover around zero, is the publication-quality figure for the reframed RQ1.
- **Mean valence trajectory by slot and condition**: a story-evolution view that does not depend on the asymmetry abstraction and helps illustrate qualitative drift.

Open additions worth considering:

- Add a per-phase semantic-exploration AUC view (the lag k axis is itself an evolution axis, but a phase split could surface fine-grained shifts).
- Replace fixed thirds with overlapping rolling windows for smoother trajectories.

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